A Structured Post-Quantum Signature Scheme Derived from a Parametric KEM Paradigm (Block-Circulant / Module-SIS Instantiation with CDT Sampler and Negacyclic NTT)

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Abstract

We present a post-quantum digital signature scheme derived from a parametric Key Encapsulation Mechanism (KEM) based on module lattices. The construction uses a seed-expanded block-circulant/module matrix, a CDT discrete Gaussian sampler, and negacyclic NTT for polynomial arithmetic. The scheme follows the Fiat–Shamir transform of an identification protocol and reduces to the hardness of the module-SIS problem. Concrete parameters targeting approximately 120-bit classical security are provided.

1 Introduction

Lattice-based cryptography is one of the most promising candidates for post-quantum security. In particular, the module-SIS problem offers a good balance between efficiency and security.

This work follows a unified paradigm:

- A KEM defines a structured hardness instance.
- A signature scheme is derived from the same structure via the Fiat–Shamir transform.

The proposed instantiation reduces structural symmetry compared to purely circulant constructions while preserving efficiency through NTT and seed-based matrix generation.

2 Preliminaries

2.1 Ring and Module Definition

Let $R_q = \mathbb{Z}_q[x]/(x^n + 1)$ where n is a power of two and q is prime. We work with module vectors:

$$A \in R_q^{k\times\ell}, \quad x \in R_q^{\ell}, \quad y \in R_q^{k}.$$

2.2 Distribution

Let D_σ denote the discrete Gaussian distribution over $\mathbb{Z}$ with standard deviation σ. Vectors are sampled as $x \leftarrow D_\sigma^{\ell}$.

3 Structured Matrix Generation

A master seed seed_A is expanded via SHAKE-256 to generate the module matrix $A \in R_q^{k\times\ell}$. Each polynomial $a_{i,j}$ is generated from an independent XOF stream. This block-circulant/module representation reduces symmetry while allowing compact representation and efficient NTT multiplication.

4 Hard Problem

Module-SIS: Given $A \in R_q^{k\times\ell}$, find a non-zero short vector $z \in R_q^\ell$ such that

$$A \cdot z = 0 \pmod q$$

with small coefficients (in absolute value).

5 Signature Scheme

5.1 Key Generation

1. Sample master seed seed_A.
2. Expand A from seed_A.
3. Sample secret $x \leftarrow D_\sigma^\ell$.
4. Compute $y = A \cdot x$.
5. Public key: (seed_A, y). Secret key: x.

5.2 Signing

Given message m:

1. Sample ephemeral $r \leftarrow D_\sigma^\ell$.
2. Compute commitment $t = A \cdot r$.
3. Compute challenge $c \leftarrow H(\mathsf{pk}\|t\|m)$ and expand to sparse polynomial of weight w.
4. Compute $z = r + c \cdot x$.
5. Reject if any coefficient of z exceeds bound B or $\|z\|_2$ exceeds threshold T.

Signature: $\sigma = (t, \mathsf{compress}(z))$.

5.3 Verification

1. Reconstruct A from seed_A.
2. Recompute $c = H(\mathsf{pk}\|t\|m)$.
3. Check $A \cdot z \stackrel{?}{=} t + c \cdot y \pmod q$.
4. Verify coefficient and norm bounds on z.

6 Correctness

$$A \cdot z = A(r + c \cdot x) = Ar + c \cdot (Ax) = t + c \cdot y.$$

7 Security Analysis

The scheme is existentially unforgeable under chosen-message attacks in the random oracle model under the hardness of module-SIS (via the forking lemma). A forger producing two signatures with the same commitment t but different challenges yields a short non-zero solution to module-SIS.

Rejection sampling ensures that the distribution of z is independent of the secret x, minimizing leakage.

8 Concrete Parameters (120-bit classical security)

Parameter	Value
n	256
q	12289
k	4
ℓ	4
σ	4.0
B	512
w (challenge weight)	60

Approximate sizes: - Public key: $\approx$ 1.6 KB - Signature: $\approx$ 5.2 KB

9 Implementation Notes

- Polynomial multiplication uses negacyclic NTT (precomputed tables). - Sampling uses CDT discrete Gaussian with domain-separated SHAKE-256. - All operations are deterministic given the seed (except ephemeral randomness).

10 Discussion and Future Work

The scheme provides a clean unified KEM–signature paradigm with reduced symmetry and good efficiency via NTT. Future work includes tighter security reductions, full constant-time implementation, and independent cryptanalysis.

11 Conclusion

We presented a structured post-quantum signature scheme derived from a parametric KEM paradigm using module-SIS. The construction is simple, scalable, and preserves a direct connection between key encapsulation and digital signatures.